

Cooperative Behaviors of Connected Autonomous Vehicles and Pedestrians to Provide Safe and Efficient Traffic in Industrial Sites*

Meng Zhang
CIAD
UBFC UTBM

Belfort Cedex, France
meng.zhang@utbm.fr

Alexandre Brunoud
CIAD
UBFC UTBM

Belfort Cedex, France
alexandre.brunoud@utbm.fr

Alexandre Lombard
CIAD
UBFC UTBM

Belfort Cedex, France
alexandre.lombard@utbm.fr

Yazan Mualla
CIAD
UBFC UTBM

Belfort Cedex, France
yazan.mualla@utbm.fr

Abdeljalil Abbas-Turki
CIAD
UBFC UTBM

Belfort Cedex, France
abdeljalil.abbas-turki@utbm.fr

Abderrafaa Koukam
CIAD
UBFC UTBM

Belfort Cedex, France
abder.koukam@utbm.fr

Abstract—The technology of Connected and Autonomous Vehicles (CAV) is a hot topic of transportation systems, especially regarding platooning and the interaction with other road users. Considering traffic safety, many studies have been devoted to the exchange of information among various road users, such as CAVs and pedestrians. In a platooning scenario, when a pedestrian is detected by a CAV, the leader CAV shares the information with its followers to provide a safe and courteous environment thanks to its connectivity. However, the possibility to improve traffic efficiency while meeting the safety requirements has rarely been addressed in current research. Yet, in industrial areas, where automated vehicles and pedestrians frequently interact, combining safety and efficiency is crucial. The present paper addresses this challenge by first analyzing the intersection of CAVs and pedestrians in no-traffic-signal scenarios. The optimal state is proposed to reduce the time loss. Then, the paper uses a reinforcement learning-based method to make CAVs arrive at the optimal state, to improve traffic efficiency. The experimental results based on virtual reality show that the proposed method increases traffic efficiency while ensuring traffic safety.

Index Terms—Intelligent Transportation System, Connected Automated Vehicle, human and robots, traffic safety

I. INTRODUCTION

With the increase of Advanced Driver Assistance Systems, the prospect of a truly autonomous vehicle has become more concrete. In response to it, a brand-new domain has emerged, called Connected and Autonomous Vehicles (CAV). This technology aims at improving the actual transportation system in terms of safety and efficiency. Indeed, researches show that CAV can reduce travel time and energy consumption [1] [2].

Autonomous and connected vehicle technologies are usually applied in closed industrial sites to ensure a stable environment. The complexity of CAV tasks can then be gradually increased. Demands for both traffic safety and traffic efficiency are highly considered. For the first consideration, many studies focus on the interaction between CAVs

and other road users, such as vulnerable pedestrians. In the scenario of interaction without traffic light between CAVs and pedestrians, safety issue is still challenging. Popular and original solutions are enhancing the information exchange between CAVs and pedestrians. The research in [3] [4] draws on the importance of pedestrian-driver interaction for traffic safety. The interactive ways proposed by many researchers are summarized in [5]. Some pedestrian behaviors are studied in [6] [7] during the crossing intersection area.

Fruitful results are archived on this topic. Based on the success of these studies, it also makes sense to consider optimizing traffic efficiency while ensuring traffic safety, especially in closed industrial areas. However, when it comes to traffic efficiency at pedestrian-vehicle intersections, the result is very limited.

In our studied scenario, pedestrians waiting in front of an intersection area (without traffic light) where are running a stream of CAVs. They interact with each other to find an appropriate crossing time for pedestrians and CAVs. During the crossing of pedestrians, CAVs control their own speed to keep a safe distance first. Then, CAVs decrease the time loss by the control algorithm under safety conditions.

To improve the traffic efficiency, we calculate the optimal state (position-speed) for CAVs during the pedestrian crossing process. The time loss can be decreased for both CAVs and pedestrians when CAVs achieve the optimal state. However, the pedestrian behavior is complex because it depends on factors (cultural, environmental, personal) as studied in [8]. Hence, it makes the control and optimization difficult. Specifically, it is hard to make CAVs reach the optimal state because of the uncertainty when pedestrians cross. To address this problem, we use a Deep Reinforcement Learning (DRL) method to train the CAV actions to reach the optimal state. The Deep Deterministic Policy Gradient (DDPG), which is a model-free off-policy algorithm for learning continuous actions, is selected to tackle the problem with the demand for continuous observations and action space. This kind

This work is supported by China Scholarship Council.

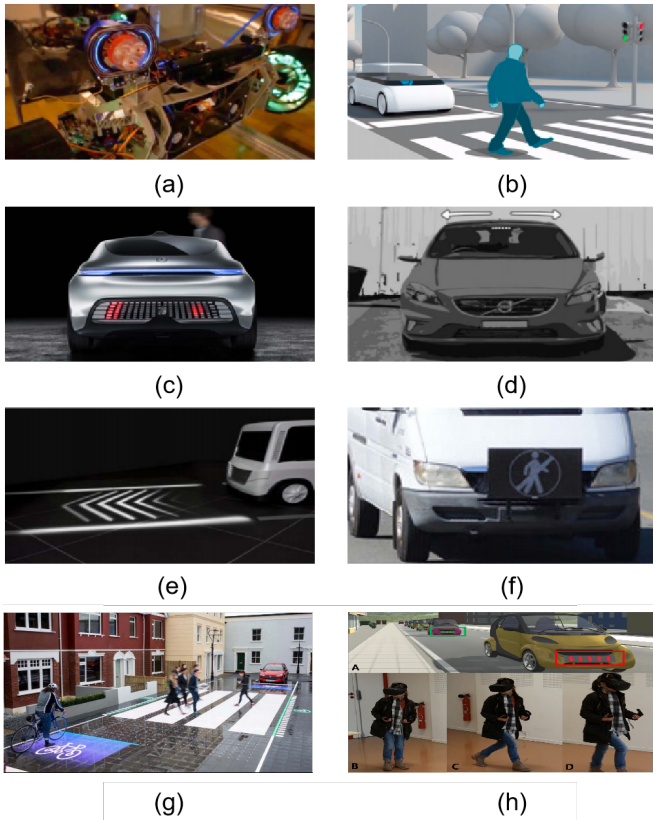


Fig. 1. Interactive behaviors between autonomous vehicles and pedestrians. (a) AEVITA moving eye concept [9] (source [10]), (b) AutoMI pedestrian detection and tracking indicator, (c) Mercedes-Benz rear-end LEDs showing that a pedestrian is crossing in front of the car, (d) an array of LEDs indicating yield [4], (e) Mitsubishi forward indicator [11], (f) an advisory display for crossing [12], (g) Examples of UMBRELLIUM smart crossing [13], and (h) Cooperative road crossing by CAV signaling system [14].

of algorithm has the advantage of dealing with long-term rewards. The reward function is built to optimize both safety and efficiency issues.

To obtain more realistic experimental results, the paper uses a new experimental method based on the Oculus Quest helmet, which is a virtual reality platform. It allows us to test our model on real pedestrian behaviors in a more realistic way. The real testers enter a virtual scene where exist a stream of running CAVs. The testers try to cross a road and interact with CAVs.

The remaining contents are organized as follows: Section II introduces the research status of vehicle-pedestrian interaction and deals with the intersection between them. Section III-A presents the model and calculates the optimal trajectory of CAV, and Section III-B constructs a DRL model to solve the control problem of CAV. Section IV designs the experiments by virtual reality simulation. Results based on DRL method, Quadratic Programming (QP), and classical way are compared and analyzed. Section V concludes the paper and introduces our future work.

II. RELATED WORK

To improve traffic safety at the intersection between autonomous vehicles and pedestrians, many researchers try to strengthen information exchange. The Work in [5] summarizes some methods and new conceptions of vehicle-

pedestrian interaction in recent years. Some popular approaches are shown in Fig. 1. In Fig. 1 (a), the AEVITA system tries to fill the information exchange by giving the autonomous vehicle the means to sense other objects in its driving environment at the first and shows the information to others. In Fig. 1 (b, d, h), autonomous vehicles show their intention to pedestrians by presenting a signal light to realize the communication between vehicles and pedestrians. In Fig. 1 (c), the vehicle prints its information to tell the following vehicle that a pedestrian is crossing. In this way, we can gain safety in the crossing pedestrian, avoiding dangerous overtaking. In Fig. 1 (e), Mitsubishi Electric Corporation provides an innovative directional-indicator system that illuminates road surfaces at night to inform pedestrians and other drivers of a vehicle's intended path forward/backward, or when turning, opening doors or making emergency stops. In Fig. 1 (f), a van representing an autonomous vehicle displays information to pedestrians informing them when to cross a street to study the interaction in different conditions. In Fig. 1 (g), the intelligent road system obtains environmental information through ground facilities, and then displays instructions on the road to improve traffic safety and efficiency.

All of the above methods and concepts are aimed at enhancing the information exchange between vehicles and pedestrians or other road users to ensure the safety of pedestrians, and fruitful results have been achieved. However, when it comes to traffic efficiency at pedestrian-vehicle intersections, there are few relevant studies. Traffic safety is the first concern, while traffic efficiency remains an important and challenging issue, especially in closed industrial areas. The research in [15] regards CAVs and pedestrians as agents and provides us with optimal speed profiles of both agents by using sequence modeling based on Petri-Net and Hamiltonian analysis. However, as can be noticed, solving the problem from the perspective of multi-agent interaction requires strong assumptions about the pedestrian behavior. Accurate estimations of pedestrian parameters ($\tilde{v}_p$: desired speed, $\bar{v}_p$: maximum speed and τ_p : reaction time) are required to support the practice. In addition, some unforeseeable events, such as the pedestrian fall, entirely question the used multi-agent optimal control approach. However, this work sets the path for the optimization of the conflict between pedestrians and CAVs, allowing gaining time for both agents as follows:

- The CAV invites earlier the pedestrian to cross by providing the soonest a reassuring safety margin.
- When the pedestrian leaves the road, the state of the CAV (its position and speed) is not only safe but also optimal to resume the journey.

In addition to these two optimization parameters, the multi-agent optimal control exhibits a kind of courtesy between both agents. The agent, who crosses the common road first, speeds up a little to free the road the soonest for the second agent. Recall that the multi-agent optimal control assumes that both agents respect their optimal trajectory. In this paper, we assume that pedestrian movements are random and all the optimization burden is performed by the CAV. To this end, DRL and QP are compared.

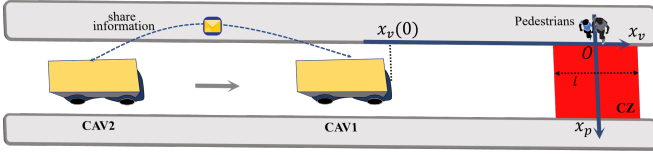


Fig. 2. Studied scenario: the pedestrian and CAVs share the road space "CZ".

III. MODELING AND CONTROL

A. Optimal trajectory of CAV

Fig. 2 presents our main idea about the intersection between CAVs and pedestrians in industrial zones. As shown, CAVs are running at their desired speed $\tilde{v}_v$ on a single lane road, while a pedestrian on the side of the road wants to cross at a moment, in an area where pedestrians are allowed to cross. It is assumed that the pedestrian's desired speed is $\tilde{v}_p$. After the detection of the pedestrian, the system creates a virtual area called "Conflict Zone" (CZ), as shown in the red rectangle. For safety reasons, CAVs and the pedestrian are forbidden to be at the same time. For CAV1, if there wasn't enough distance to brake to let the pedestrians cross first, CAV1 shows a red light, as shown in Fig. 1 (h), to tell pedestrians to wait, and CAV1 share this message to CAV2. In this way, CAV2 can prepare itself as early as possible to yield the way to pedestrians. The objective is to control CAVs' longitudinal speed to improve traffic safety and efficiency. Precisely, it is to avoid collision and decrease the time lost by CAVs and pedestrians during the crossing.

For the traffic safety, the technology used in Fig. 1 (g) can guarantee information exchange between road users. And the speed of CAV is limited according to its distance to CZ before pedestrians exit, as expressed in equation (1) where x_v , u_v , v_v are CAV position, acceleration, and speed respectively. This means that the actual distance to CZ is no less than the shortest braking distance plus a margin v_v multiplying a positive gain τ .

$$|x_v(t) - \frac{l}{2}| \geq \frac{-v_v^2(t)}{2 \cdot u_v} + \tau v_v(t) \quad (1)$$

Speed v_v and acceleration u_v constraints are considered as follows:

$$u_v(t) - \bar{u}_v \leq 0 \quad (2)$$

$$\underline{u}_v - u_v(t) \leq 0 \quad (3)$$

$$v_v(t) - \bar{v}_v \leq 0 \quad (4)$$

$$0 - v_v(t) \leq 0 \quad (5)$$

Where $\bar{u}_v$ and $\underline{u}_v$ designate the maximum and the minimum acceleration (deceleration), respectively. The maximum speed is $\bar{v}_v$.

After having the above definitions, we can analyze the efficiency in the case of pedestrians crossing first. Because the CAV maintains a safe distance to the pedestrian when she/he is in CZ (equation (1)), there is an optimal speed and gap that let the CAV be the furthest from CZ after the pedestrian exit. The CAV state before pedestrians exit CZ is defined as $S = (x_v, v_v)$. The optimal state S^* is the state that CAV can get the biggest position when it accelerates to the desired speed $\tilde{v}_v$.

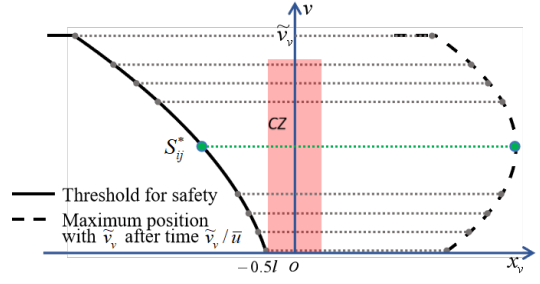


Fig. 3. The relationship between S and the position after time $\frac{\tilde{v}_v}{\bar{u}}$ when CAVs, with all possible S , can reach the desired speed $\tilde{v}_v$. The CAV with optimal state S^* will run the furthest.

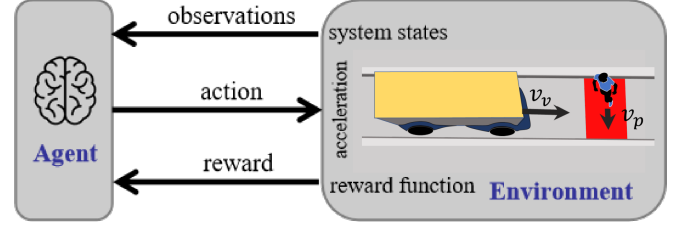


Fig. 4. DRL architecture for the problem of intersection between a CAV and a pedestrian.

To illustrate the concept, Fig. 3 reveals the relationship between the safe gap and the final position of CAV. The optimal state S^* can be expressed by expression (6), where variables are simplified in writing.

$$S^* = \left(\frac{(\bar{u}^2 - 2\bar{u}\underline{u} + 2\underline{u}\tilde{v}_v - \tilde{v}_v^2)\underline{u}}{2(\bar{u} - \underline{u})^2} - \frac{l}{2}, \frac{(\tau\bar{u} - \tilde{v})\underline{u}}{\bar{u} - \underline{u}} \right) \quad (6)$$

B. DRL model

1) *Network and observations*: Fig. 4 shows us the process of dealing with the intersection problem by using DRL in our studied scenario. The cycle begins with the Agent observing the Environment and receiving a state and a reward. The Agent uses this state and reward for deciding the next action to take. The Agent then sends an action to the Environment in an attempt to control it favorably. Finally, the environment transitions, and its internal state changes as a consequence due to the previous state and the Agent's action. Then, the cycle repeats. Specifically, the action is the acceleration applied to the CAV. The observations are information about the CAV and the pedestrian.

In this article, DDPG agent is selected to tackle the problem with the demand for continuous observations and action space. DDPG is an off-policy method and entirely model-free. Detailed information about the DDPG agent and its variables can be found in [16]. For the structure of networks, we refer to and adjust the structure presented in Fig. 3 in [17], which is proved to be effective in car following. The observations are ΔL (see equation (7)), $x_v(t)$, $v_v(t)$, $v_v(t) - \tilde{v}_v$, $\int (v_v(t) - \tilde{v}_v)^2 dt$, $x_p(t)$, $v_p(t)$, and $f_p(t)$ ($f_p(t) = 1$ when the pedestrian is in CZ, otherwise 0).

2) *Reward function*: To train the DDPG agent with the ability to reach state S^* when pedestrians exit the road, a reward function, given in expression (7) is defined. In this function, reward r_1 represents a reward for safety constraint, equation (1). If the CAV state fulfills the safety constraints

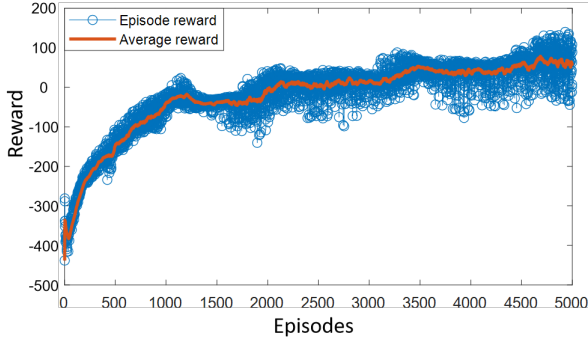


Fig. 5. The training curves of the episode reward and average reward.

($\Delta L \geq 0$), it gives a positive reward ($0.01 \cdot w_1$) to the agent, otherwise a negative reward. r_2 is the final position (when $t = t_{final}$, the time when CAV reaches desired speed) that the CAV can reach. It evaluates the state of CAVs when pedestrians exit their respective lanes. The reward r_2 is higher when the state of CAVs is closer to optimal state S^* . Therefore, this reward guides the agent to reach the optimal state during the training. r_3 is to punish the agent if CAVs exceed the speed $\bar{v}_v$. In r_4 , u_{t-1} is the acceleration from the previous time step. Finally, r_4 is to decrease the control effort. Finally, the full DDPG learning process can be found in [16].

$$\begin{aligned}
 r(t) &= w_1 \cdot r_1 + w_2 \cdot r_2 - w_3 \cdot r_3 - w_4 \cdot r_4 \\
 r_1 &= 0.01 (\Delta L \geq 0) + \Delta L (\Delta L < 0); \\
 r_2 &= \cdot x_v(t_{final}), (t = t_{p,exit}); \\
 r_3 &= v_v(t) - \bar{v}_v, (v_v(t) > \bar{v}_v); \\
 r_4 &= u_{t-1}^2, (t \leq t_{p,exit}); \\
 \Delta L &= \cdot x_v(t) - \frac{l}{2} - \left(\frac{v_v^2(t)}{-2 \cdot \underline{u}} + \tau v_v(t) \right)
 \end{aligned} \tag{7}$$

IV. EXPERIMENTS AND RESULTS

A. Training setting

For the environment, The DDPG agent is realized through MATLAB, and the DRL environment is built through SIMULINK. Weighting factors in expression (7) are set to: $w_1 = 50$, $w_2 = 2$, $w_3 = 10$, $w_4 = 0.01$. Other parameters are: $\tilde{v}_v = 10$, $\bar{u} = 2$, $\underline{u} = -3$, $\bar{v}_v = 1.2\tilde{v}_v$, $\tau = 1$, road width $4m$. According to (6) and the studied scenario parameters, the optimal state S^* is $(-13.64m, 4.8m/s)$ according to expression (6). The cooperative control area is set to $(-100, 0)$. And the crossing speed of pedestrians is set to $(0, 2)m/s$. We give a possibility of 0.2 to let pedestrians stop in CZ. In this way, we can train the agent to deal with dangerous situations. When launching a training episode, we randomly initialize the positions of CAVs in the defined cooperative control area, and they drive normally at the desired speed. Fig. 5 shows the training curve, and it converges well.

B. Experiments and results

The purpose of these experiments is to test if the developed algorithm is applied to the behaviors of real pedestrians and the improvement of traffic efficiency. Experiments have been conducted with a mixed reality platform. The testers play the

role of pedestrians and are asked to physically cross a road in a virtual environment provided by the Oculus Quest. This device ensures safety during the experiments and provides immersive scenes to be as close as possible to the real world. In all the experiments, testers interact with a stream of unmanned vehicles: no visible driver on the virtual vehicle.

1) *Comparison of DRL and QP*: To test the effectiveness of trained agent and its advantages, we compare CAVs behaviors under the controller and QP. For the design of the QP controller, we use the same speed and acceleration constraints for CAVs, then estimate the pedestrian crossing time, and plan the CAV state to the optimal state S^* when the pedestrians leave the CZ. We compute a QP solution on the pedestrian test data (at each step) and QP outputs the value of CAV acceleration.

The result as shown in Fig. 6 (e), we see the classic pattern of a pedestrian crossing: looking and analyzing the CAV trajectory, starting to cross while looking at the CAV (ensuring safety and understanding with the CAV) and finally crossing without looking (clearing the path) [3].

We then analyze the position, speed, and acceleration shape from both solutions and check whether the safety constraint has been respected. In Fig. 6 (a), the CAV provides an earlier reduced speed to invite the pedestrian to cross as soon as possible. Indeed, DRL method can anticipate pedestrian behavior whereas QP is only using the actual pedestrian features. In the same way, CAV is leaving the CZ earlier and with more speed than QP solution. We can notice in Fig. 6 (b) that the safety constraint is respected as long as the pedestrian is not leaving the crosswalk. Moreover, the DRL acceleration shape is smoother. Therefore, CAV shows impressive results with great efficiency (better than QP solution) while maintaining the safety constraint.

2) *Multiple lanes and CAVs*: In this experiment, we test our algorithm in a more complex environment where are two lanes and 4 CAVs. The scenario and result are shown as in Fig. 7. All the CAV act at the same time to communicate a safe trajectory to the pedestrian. This test also shows all the possible actions of the CAV: yield or drive through for the first and second lanes. When the car is too close, it accelerates (with a red signal) to shorten the pedestrian waiting time. When the car can stop, a green signal is displayed and CAV is slowing down until the pedestrian leaves the crosswalk. The test is conclusive: the CAVs are close to the optimal state before the pedestrian exit, which ensures their safety while performing well.

3) *Comparison of DRL and classical way*: To highlight the advantages of the proposed approach in terms of efficiency, a comparison is performed with the classical slow-down behavior. We did the test with 28 real testers. They did the experiment 2 times. In the first, CAVs without signal light keep its maximal speed under safety conditions as described in Fig. 3. In the second, CAVs with a colored signal light use DRL controller. In all scenarios, pedestrians start crossing when they feel safe.

In the spontaneous scenario, we observe two ranges of pedestrians. The first range (35.71%) forces the CAV to stop using hesitating movements, wait for the complete stop of the CAV and then slowly cross the street. The second range of pedestrian (64.28%) is using the rolling gap strategy [8].

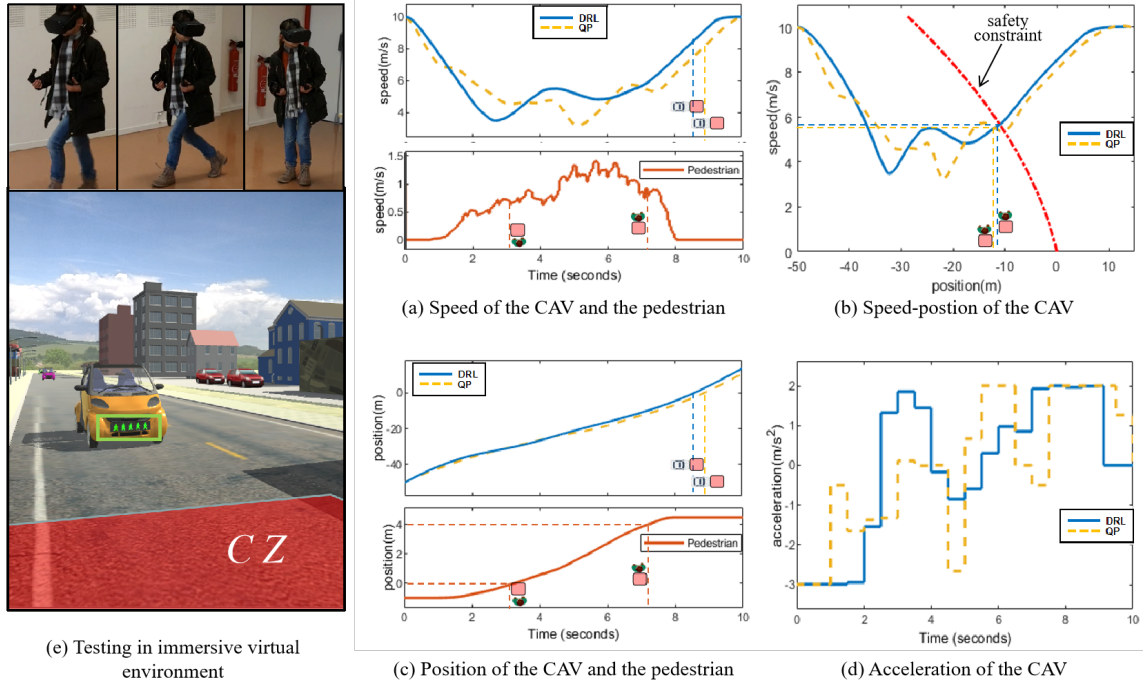


Fig. 6. A real person participates in the game by using an immersive hamlet technology.

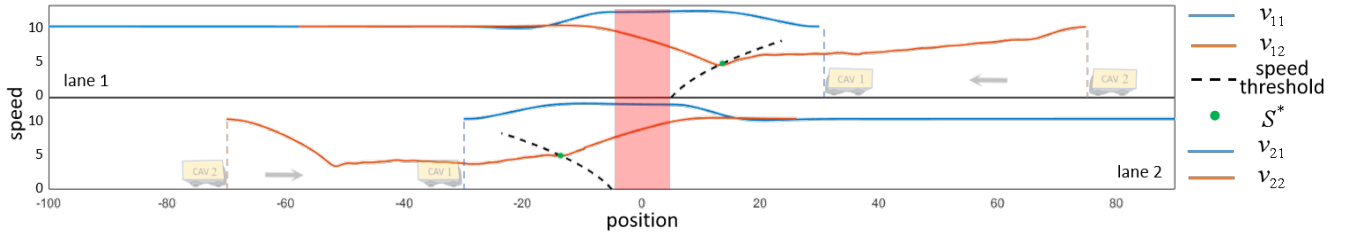


Fig. 7. CAVs speed profiles with 2 lanes and 4 CAVs: The first CAVs in all lanes accelerate to $\bar{v}_v$ and the following CAVs controlled by DRL slow down at the same time. After the first CAVs pass through CZ, pedestrians start crossing. When the pedestrians exit the lane, the state of the CAV in this lane is close to S^* . In the results, the average pedestrian waiting time is 3.6s ($STD : 0.45$). The average CAV positions are 27.1 ($STD : 1.59$) and -27.9 ($STD : 1.90$) after 5s of pedestrian exiting their lane.

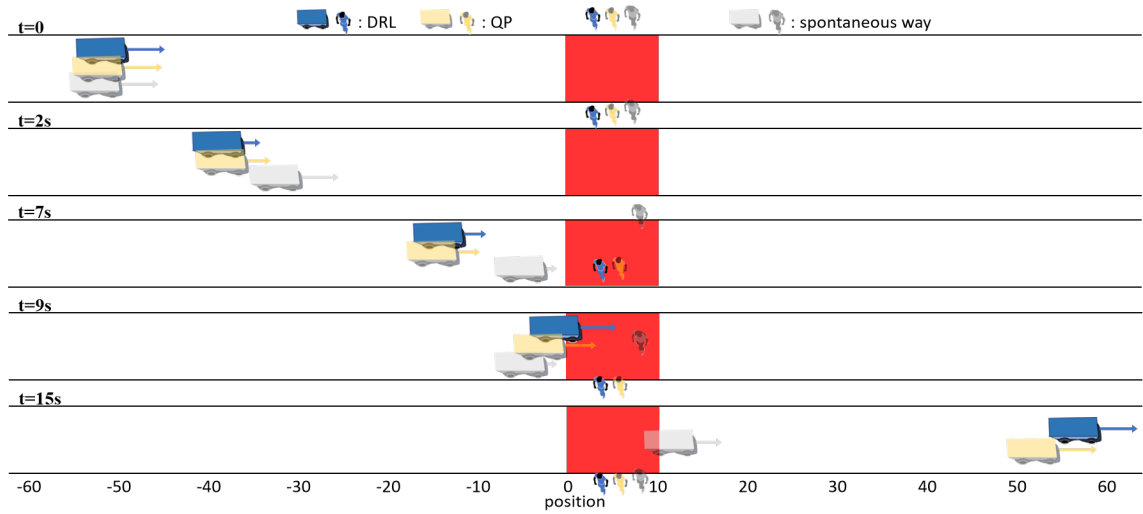


Fig. 8. A comparison between three scenarios: DRL method, QP solution, and spontaneous way. The arrows represent CAV speed.

Averagely, the pedestrian crossing delays the CAV by $6.21s \pm 3.03s$ in the spontaneous scenario, whereas in the DRL scenario, the delay equals $3.85s \pm 2.20s$. The DRL agent allows the CAV gaining averagely $2.36s$. The crossing of CAV delays pedestrian by $5.25s \pm 1.03s$ in the spontaneous scenario, whereas in the DRL scenario, the delay equals $3.53s \pm 1.60s$. The DRL agent allows the pedestrian gaining averagely $1.72s$. The big value of the standard deviation is because there are 6 crossings among the 28 with a low gain. More precisely, there are 4 crossings where the CAV was initially far from CZ and 2 crossings where the pedestrian decision-making was only based on the displayed green without waiting for the CAV beginning the slowdown process.

V. CONCLUSION

The present work focuses on the industrial environment where pedestrians are trained, and the crossing perimeters are well delimited. In this context, the obtained results show that this work deserves to be extended. In terms of optimization, it is interesting to extend the approach to several lanes (more than two lanes) and several crossing zones. From the human-machine interaction, connected wearable devices, such as vibrating safety jackets, and ground signalization are worth to be studied for improving both safety and efficiency. Finally, in a more general context, as in urban areas, the results invite us to study the speed profile as a vector of communication with the other road users, in addition to the other studied signaling systems.

REFERENCES

- [3] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Agreeing to cross: How drivers and pedestrians communicate," *CoRR*, vol. abs/1702.03555, 2017. [Online]. Available: <http://arxiv.org/abs/1702.03555>
- [4] T. Lagström and V. Malmsten Lundgren, "Avip-autonomous vehicles' interaction with pedestrians: an investigation of pedestrian-driver communication and development of a vehicle external interface," Master's thesis, 2016.
- [5] A. Rasouli and J. K. Tsotsos, "Autonomous vehicles that interact with pedestrians: A survey of theory and practice," *IEEE transactions on intelligent transportation systems*, vol. 21, no. 3, pp. 900–918, 2019.
- [6] R. Tian, E. Y. Du, K. Yang, P. Jiang, F. Jiang, Y. Chen, R. Sherony, and H. Takahashi, "Pilot study on pedestrian step frequency in naturalistic driving environment," in *2013 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2013, pp. 1215–1220.
- [7] R. Tian, L. Li, K. Yang, S. Chien, Y. Chen, and R. Sherony, "Estimation of the vehicle-pedestrian encounter/conflict risk on the road based on taxi 110-car naturalistic driving data collection," in *2014 IEEE Intelligent Vehicles Symposium Proceedings*. IEEE, 2014, pp. 623–629.
- [8] N. M. Zafri, R. Sultana, M. R. H. Hima, and T. Tabassum, "Factors influencing pedestrians' decision to cross the road by risky rolling gap crossing strategy at intersections in dhaka, bangladesh," *Accident Analysis & Prevention*, vol. 142, p. 105564, 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0001457519318615>
- [9] N. Pennycooke, "Aevita: Designing biomimetic vehicle-to-pedestrian communication protocols for autonomously operating & parking on-road electric vehicles," Ph.D. dissertation, Massachusetts Institute of Technology, 2012.
- [10] B. Färber, "Communication and communication problems between autonomous vehicles and human drivers," in *Autonomous driving*. Springer, 2016, pp. 125–144.
- [11] M. Electric, "Mitsubishi electric introduces road-illuminating directional indicators." 2015. [Online]. Available: <https://www.mitsubishielectric.com/news/2015/1023.html>
- [12] M. Clamann, M. Aubert, and M. L. Cummings, "Evaluation of vehicle-to-pedestrian communication displays for autonomous vehicles," Tech. Rep., 2017.
- [13] Umbrellium, "Starling crossing is an interactive pedestrian crossing that responds dynamically in real-time to make pedestrians, cyclists & drivers safer and more aware of each other." 2020. [Online]. Available: <https://umbrellium.co.uk/projects/starling-crossing/>
- [14] M. Zhang, A. Abbas-Turki, A. Lombard, and A. Koukam, "Connected and autonomous vehicles cooperate with the pedestrian in industrial sites based on trajectory optimization and vehicle signalization system," in *2020 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2020, pp. 188–194.
- [15] M. Zhang, A. Abbas-Turki, A. Lombard, A. Koukam, and K.-H. Jo, "Autonomous vehicle with communicative driving for pedestrian crossing: Trajectory optimization," in *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2020, pp. 1–6.
- [16] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," *arXiv preprint arXiv:1509.02971*, 2015.
- [17] Y. Ye, X. Zhang, and J. Sun, "Automated vehicle's behavior decision making using deep reinforcement learning and high-fidelity simulation environment," *Transportation Research Part C: Emerging Technologies*, vol. 107, pp. 155–170, 2019.